

# Vitamin D

Fact Sheet for Health Professionals

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For information on vitamin D and COVID-19, see [\*Dietary Supplements in the Time of COVID-19\*](#).

## Introduction

Vitamin D (also referred to as calciferol) is a fat-soluble vitamin that is naturally present in a few foods, added to others, and available as a dietary supplement. It is also produced endogenously when ultraviolet (UV) rays from sunlight strike the skin and trigger vitamin D synthesis.

Vitamin D obtained from sun exposure, foods, and supplements is biologically inert and must undergo two hydroxylations in the body for activation. The first hydroxylation, which occurs in the liver, converts vitamin D to 25-hydroxyvitamin D [25(OH)D], also known as calcidiol. The second hydroxylation occurs primarily in the kidney and forms the physiologically active 1,25-dihydroxyvitamin D [1,25(OH)2D], also known as calcitriol [1].

Vitamin D promotes calcium absorption in the gut and maintains adequate serum calcium and phosphate concentrations to enable normal bone mineralization and to prevent hypocalcemic tetany (involuntary contraction of muscles, leading to cramps and spasms). It is also needed for bone growth and bone remodeling by osteoblasts and osteoclasts [1-3]. Without sufficient vitamin D, bones can become thin, brittle, or misshapen. Vitamin D sufficiency prevents rickets in children and osteomalacia in adults. Together with calcium, vitamin D also helps protect older adults from osteoporosis.

Vitamin D has other roles in the body, including reduction of inflammation as well as modulation of such processes as cell growth, neuromuscular and immune function, and glucose metabolism [1-3]. Many genes encoding proteins that regulate cell proliferation, differentiation, and apoptosis are modulated in part by vitamin D. Many tissues have vitamin D receptors, and some convert 25(OH)D to 1,25(OH)2D.

In foods and dietary supplements, vitamin D has two main forms, D<sub>2</sub> (ergocalciferol) and D<sub>3</sub> (cholecalciferol), that differ chemically only in their side-chain structures. Both forms are well absorbed in the small intestine. Absorption occurs by simple passive diffusion and by a mechanism that involves intestinal membrane carrier proteins [4]. The concurrent presence of fat in the gut enhances vitamin D absorption, but some vitamin D is absorbed even without dietary fat. Neither aging nor obesity alters vitamin D absorption from the gut [4].

## Assessing vitamin D status

Serum concentration of 25(OH)D is the main indicator of vitamin D status. However, the serum concentrations of 25(OH)D that are associated with vitamin D deficiency have not been definitively identified. The Food and Nutrition Board at the National Academies of Sciences, Engineering, and

Medicine states that levels of 50 nmol/L (20 ng/mL) or more are sufficient for most people, and that the risk of deficiency increases at serum concentrations of less than 30 nmol/L (12 ng/mL).

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## Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for vitamin D. These values range from 15 to 20 mcg (600–800 IU) for adults and from 10 to 15 mcg (400–600 IU) for infants, children, and adolescents, depending on age.

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## Sources of Vitamin D

### Food

Vitamin D is found naturally in a few foods, such as the flesh of fatty fish, fish liver oils, beef liver, egg yolks, and cheese. In American diets, fortified foods (e.g., milk, breakfast cereals) provide most of the vitamin D.

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### Sun exposure

People can get some of their daily vitamin D through exposure to sunlight, although the season, time of day, length of day, cloud cover, melanin content of the skin, and other factors can affect ultraviolet radiation exposure and vitamin D synthesis.

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### Dietary supplements

Vitamin D is present in dietary supplements as either vitamin D<sub>2</sub> or vitamin D<sub>3</sub>. Both can raise the serum level of 25(OH)D. However, research shows that vitamin D<sub>3</sub> increases serum 25(OH)D levels to a greater extent than vitamin D<sub>2</sub> and can maintain those higher levels for longer periods of time.

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## Vitamin D Intakes and Status

According to data from the National Health and Nutrition Examination Survey, most people in the United States consume less than the recommended amounts of vitamin D. However, evidence suggests that the majority of people have sufficient serum concentrations of vitamin D based on the thresholds set by the Food and Nutrition Board. Sun exposure is one of the reasons why serum levels of 25(OH)D are generally higher than would be predicted on the basis of vitamin D dietary intakes alone.

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# Vitamin D Deficiency

Vitamin D deficiency is more common among people who are lactose intolerant, have a milk allergy, or follow an ovo-vegetarian or vegan diet. Deficiency may also occur in people who have limited exposure to sunlight, those whose kidneys cannot convert 25(OH)D to its active form, or those who cannot absorb vitamin D efficiently in the digestive tract. Vitamin D deficiency can manifest as rickets in children and as osteomalacia in adolescents and adults.

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## Groups at Risk of Vitamin D Inadequacy

Certain groups of people are more likely than others to have inadequate vitamin D status. These include breastfed infants, older adults, people with limited sun exposure, people with dark skin, people with conditions that limit fat absorption, and people with obesity or those who have undergone gastric bypass surgery.

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## Vitamin D and Health

The FNB committee that established DRIs for vitamin D found that the evidence was inadequate or too contradictory to conclude that the vitamin had any effect on a long list of potential health outcomes (e.g., on resistance to chronic diseases or functional measures), except for measures related to bone health. Similarly, in a review of data from nearly 250 studies published between 2009 and 2013, the Agency for Healthcare Research and Quality concluded that no relationship could be firmly established between vitamin D and health outcomes other than bone health [69]. However, because research has been conducted on vitamin D and numerous health outcomes, this section focuses on seven diseases, conditions, and interventions in which vitamin D might be involved: bone health and osteoporosis, cancer, cardiovascular disease (CVD), depression, multiple sclerosis (MS), type 2 diabetes, and weight loss.

Most of the studies described in this section measured serum 25(OH)D levels using various methods that were not standardized by comparing them to the best methods. Use of unstandardized 25(OH)D measures can raise questions about the accuracy of the results and about the validity of conclusions drawn from studies that use such measures and, especially, from meta-analyses that pool data from many studies that use different unstandardized measures [5,9,70]. More information about assay standardization is available from the [Vitamin D Standardization Program](#) webpage.

### Bone health and osteoporosis

Osteoporosis is characterized by low bone mass and the deterioration of bone tissue, which increases bone fragility and the risk of fractures. Clinical trials have shown that vitamin D and calcium supplements may increase bone mineral density in some postmenopausal women and older men, but it is not clear whether they reduce falls and fracture rates. In addition, the results of

studies that have evaluated the effects of supplemental vitamin D on muscle strength and the rate of decline in muscle function have been inconsistent.

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## **Cancer**

Some evidence suggests that vitamin D might inhibit carcinogenesis and slow tumor progression. However, most research has found no relationship between serum 25(OH)D levels and risk of cancer. The results of clinical trials have generally failed to show that vitamin D supplementation with or without calcium supplementation reduces the incidence of cancer. Adequate or higher 25(OH)D levels might reduce cancer mortality rates, but more research is needed to determine the effects of vitamin D supplementation in people with cancer.

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## **Cardiovascular disease**

Vitamin D has been linked to heart health and the risk of cardiovascular disease and vitamin D deficiency is associated with vascular dysfunction, arterial stiffening, left ventricular hypertrophy, and hyperlipidemia. While data from observational studies support an association between higher serum levels of 25(OH)D and a lower risk of cardiovascular disease incidence and mortality, clinical trials have not confirmed these results; several trials have reported that vitamin D supplementation did not reduce the risk of cardiovascular disease, even among people with low 25(OH)D status.

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## **Depression**

Vitamin D may be involved in the pathophysiology of depression, and low levels of 25(OH)D have been associated with a higher risk of depression. However, vitamin D supplementation has not been shown to prevent depression or treat depressive symptoms in clinical trials.

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## **Multiple sclerosis**

Epidemiological and genetic studies have found an association between multiple sclerosis and low 25(OH)D levels. Observational studies suggest that adequate vitamin D concentrations might reduce the risk of contracting multiple sclerosis, slow the disease's progression, and decrease the risk of relapse. Clinical trial evidence is limited, so experts cannot draw conclusions about whether vitamin D can help prevent multiple sclerosis, but vitamin D supplementation does not appear to help mitigate the signs and symptoms of the disease or reduce relapse rates.

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## **Type 2 diabetes**

Observational studies have found an inverse association between vitamin D status and blood glucose levels. However, vitamin D supplementation does not appear to help maintain glucose

homeostasis, reduce the risk of progression from prediabetes to type 2 diabetes, or help manage the disease, particularly in vitamin D-replete individuals.

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## Weight loss

Although there is some evidence that higher body weights are associated with lower vitamin D status, the available research does not support the use of vitamin D supplements to promote weight loss.

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## Health Risks from Excessive Vitamin D

Vitamin D toxicity can cause hypercalcemia, hypercalciuria, and high serum 25(OH)D concentrations; in extreme cases, it may lead to renal failure, calcification of soft tissues, cardiac arrhythmias, and death. Vitamin D toxicity is almost always a result of excessive intakes of vitamin D through supplements. Taking calcium supplements in combination with vitamin D supplements may increase the risk of certain adverse effects. The Tolerable Upper Intake Level for vitamin D ranges from 25 to 100 mcg (1,000–4,000 IU), depending on age.

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## Interactions with Medications

Vitamin D supplements may interact with medications, and some medications may affect vitamin D levels. These medications include orlistat, statins, steroids, and thiazide diuretics.

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## Vitamin D and Healthful Diets

In general, a person's nutritional needs should be met primarily through foods. Fortified foods and dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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